WELLSLING · PRODUCT REQUIREMENTS AND SPECIFICATIONS

WellSling product requirements and specifications

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	Replaces WSL-REQ-001 v0.2 (kit 1.9.1 upgrade). R1 to R13 migrated to PR-001 to PR-013 with the same targets and verification, tolerance, units and traced user requirements added; PR-014 to PR-017 added from the constraints and settled questions; status from WSL-CAL-001 v0.2

WellSling product requirements and specifications

What the design must do, each traced to a user need in WSL-URS-001. Fourteen of the seventeen requirements are met on paper or by design. PR-006 (setup in 20 minutes) is at risk: the estimate meets the target with no margin. PR-009 is reported against the value-engineering target, which the constructable design exceeds by USD 1,593. PR-008 (operation after a short briefing) can only be shown in a user trial. Results come from WSL-CAL-001 v0.2; tags in square brackets point to its sections. Targets are proposals for review with a co-design partner. Method: STANDARDS section 22.

> **Safety:** WellSling lifts a heavy, frightened animal over an open well. Nobody goes down the well at any stage. The kit is an open engineering reference, not certified lifting or rescue equipment, and it must be proof-loaded before any use (TRL 4).

Table 1. Product requirements and specifications.

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
PR-001	UR-001	Safety	No entry	Bands shall be passed, rigged, lifted and landed entirely from the rim	0	max	people below the rim	Inspection of the method (BLD-001), TRL 3; trial with a weighted dummy in a test pit, TRL 4	Met by design	R1
PR-002	UR-002, UR-011	Performance	Lifting capacity	The kit shall lift the design animal with the hook line leaning up to 10 deg, every steel part at a factor of 3 on yield and buckling and the rope at 5, and shall be proof-loaded to 1.5 times	600 (662 at the hook)	min	kg	Analysis [A2, B1, B10, C5, C6], TRL 3; proof-load test, TRL 4	Met on paper (static; the dynamic load of a kicking animal to measure at TRL 4)	R2
PR-003	UR-008	Performance	Band spreads the load	Each band shall be at least 200 mm wide, bear on a straight bar at its eyes, and the average pressure on the animal shall be stated	200	min	mm width	Analysis [E6, E7], TRL 3; design review with a veterinary partner, TRL 4	Met on paper for width: two 240 mm bands, about 17 to 21 kPa; tolerance to verify at TRL 4	R3
PR-004	UR-003	Performance	Passer reach	The passer shall reach an animal 15 m below the rim	15	min	m	Analysis [G3], TRL 3; trial in a test pit, TRL 4	Met on paper: 16.5 m	R4
PR-005	UR-004	Physical architecture	Well span	The kit shall straddle wells up to 3	3.0	min	m	Analysis [C9, F4], TRL 3; fit	Met: pads 700 mm,	R5

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
				m across with every pad at least 650 mm and every bearer at least 350 mm back from the rim				check on sample wells, TRL 4	bearers 400 mm back	
PR-006	UR-005	Usability	Fast setup	Three people shall have the kit ready to lift within 20 minutes of arrival	20	max	min	Analysis [G4], TRL 3; timed drill, TRL 4	At risk: 20 min estimate with no margin	R6
PR-007	UR-007	Safety	Controlled lift	The winch brake shall hold the load at any point when the handle is released	100	min	% of hook load held	Inspection of the winch specified, TRL 3; brake hold test at proof load, TRL 4	Met by selection of a load-pressure brake	R7
PR-008	UR-006	Usability	Simple operation	First-time operators shall complete a lift after a briefing	15	max	min of briefing	User trial with first-time operators, TRL 4	To verify at TRL 4	R8
PR-009	UR-014	Cost	Value-engineering target	The whole kit shall be costed against the value-engineering target, a hypothetical control target and not a limit	1,500	target	USD	Costed BOM [G2], TRL 3	Over target: USD 3,093, USD 1,593 over	R9
PR-010	UR-011	Safety	Tripod stability	No foot shall lift with the hook line leaning 10 deg in any direction	10	min	deg	Analysis [C3, C4], TRL 3; lean test at proof load, TRL 4	Met on paper: tips only at 19.4 deg	R10

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
PR-011	UR-010	Usability	Hand effort	Crank force shall stay at or below the target throughout the lift	250	max	N	Analysis [B6], TRL 3; force gauge on the handle, TRL 4	Met on paper: 200 N at most, on an assumed handle force	R11
PR-012	UR-009	Physical architecture	Landing clearance	Hooves shall be at least 150 mm above the deck at the landing height so the bridge slides under	150	min	mm	Analysis [F5], TRL 3; fit check with a dummy, TRL 4	Met on paper: 169 mm	R12
PR-013	UR-013	Packaging and transport	Carry weight	No single piece shall be heavier than two people can carry	35	max	kg	Analysis [G1], TRL 3; weigh the parts, TRL 4	Met on paper: 32.0 kg	R13
PR-014	UR-012	Safety	Rim restraint	Anyone within 1 m of the rim shall wear a harness on a work-restraint lanyard tied to a foot and set so the wearer cannot reach the edge	1.0	min	m from the rim	Inspection of the kit and method (BOM line 28, BLD-001), TRL 3; drill, TRL 4	Met by design	
PR-015	UR-015	Materials	Outdoor life	Every steel part shall be hot-dip galvanised after welding, with vent holes in closed sections	70	min	um mean zinc on 3 to 6 mm steel	Inspection of the BOM and analysis (STEM S28), TRL 3; coating gauge, TRL 4	Met by design	
PR-016	UR-016	Materials			100	min				

ID	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status	Was
			Local making	Every made part shall be cut, drilled, bent or welded from plate, tube, box section, bar or timber with workshop tools			% of made parts	Inspection of the drawings and build plan (BLD-001), TRL 3	Met by design	
PR-017	UR-017	Physical architecture	Design-arounds	The bands shall be plain webbing with no stiffening insert, and the passer a separate tool never fixed to a band	0	max	stiffening inserts	Inspection of the BOM (lines 18 to 22), TRL 3	Met by design	

Categories: physical architecture, performance, materials, usability, packaging and transport, safety, electrical, software. Status: met on paper, met by design, at risk, not met, over target, withdrawn, to verify at TRL 4.

Assumptions

- The design animal is 600 kg, the upper end of a grown cow or buffalo; a heavier animal is outside the rating.
- The ground around the well bears at least 75 kPa under a timber pad; a softer site needs larger pads.
- A plain webbing band with a loose felt sleeve can lift a large animal for the length of a lift without injury when two bands share the load. This is to be confirmed with a veterinary partner.
- Most open wells where animals fall are 3 m across or less.
- A village or cooperative keeps the kit in one known place and practises with it.